

CHUBU CENTRAIR INTL, Japan

WMO: 476345

Lat: 34.858N

Lon: 136.805E

Elev: 5

StdP: 101.27

Time Zone: 9.00 (E09)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.0	2.0	-6.8	2.1	4.3	-5.2	2.4	4.6	17.2	5.5	15.7	5.1	6.9	350	0.539

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.1	32.1	26.3	31.1	26.0	30.1	25.6	27.1	30.3	26.6	29.7	26.2	29.1	5.0	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.1	21.5	29.5	25.8	21.2	29.2	25.1	20.2	28.5	85.6	30.9	83.5	29.8	81.4	29.5	30.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
15.1	13.5	12.2	DB	-0.6	34.1	1.4	1.0	-1.6	34.8	-2.5	35.4	-3.2	35.9	-4.3	36.7	
			WB	-2.7	28.2	0.9	0.9	-3.3	28.8	-3.8	29.3	-4.3	29.8	-4.9	30.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.6	6.1	6.5	9.3	14.0	18.9	22.7	26.5	28.1	24.7	19.6	13.9	8.7
	DBStd	7.91	1.95	2.29	2.55	2.65	2.08	2.15	2.07	1.69	2.36	2.39	2.54	2.53
	HDD10.0	328	123	101	44	2	0	0	0	0	0	0	2	56
	HDD18.3	1584	381	332	279	130	17	1	0	0	0	13	133	299
	CDD10.0	2752	1	3	23	124	277	379	512	560	441	298	119	16
	CDD18.3	967	0	0	0	1	37	130	254	302	191	52	1	0
	CDH23.3	7546	0	0	0	1	52	466	2269	3309	1347	102	0	0
	CDH26.7	2086	0	0	0	0	2	67	621	1139	253	4	0	0
Wind	WSAvg	5.6	7.0	6.8	6.3	5.6	5.1	4.3	4.2	4.4	4.9	5.3	5.7	7.0
Precipitation	PrecAvg	1673	56	70	111	136	173	209	181	144	275	163	98	57
	PrecMax	2209	167	144	208	234	385	302	363	404	544	512	264	121
	PrecMin	1007	7	28	48	38	82	77	42	25	49	24	8	4
	PrecStd	293	41	32	41	59	76	63	91	91	128	93	65	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.2	14.8	18.1	22.2	26.2	30.9	33.0	33.9	31.9	27.0	21.1	17.1
		MCWB	8.3	11.9	12.7	16.6	19.5	25.2	26.8	26.8	26.1	22.1	16.5	14.0
	2%	DB	11.1	12.2	16.1	20.8	24.9	28.2	31.9	32.2	30.1	25.8	20.0	15.1
		MCWB	7.5	9.1	12.2	15.6	19.0	23.4	26.4	26.3	25.1	21.1	15.6	11.6
	5%	DB	10.1	11.2	14.9	19.1	23.2	27.0	30.9	31.8	29.1	24.2	19.0	14.0
		MCWB	6.8	8.1	11.5	15.0	18.0	22.7	26.1	26.2	24.5	20.1	14.9	10.4
	10%	DB	9.2	10.2	13.2	18.1	22.2	25.9	29.9	30.9	28.1	23.2	18.1	12.9
		MCWB	5.9	7.2	9.9	14.4	17.6	22.1	25.6	25.9	23.9	19.0	14.0	9.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.9	12.6	14.7	18.2	21.5	25.8	27.9	28.0	26.9	23.6	18.4	15.1
		MCDB	11.2	14.2	16.4	20.4	24.3	29.2	31.5	31.5	30.2	25.5	19.8	16.3
	2%	WB	8.2	10.4	13.4	17.2	20.3	24.7	26.9	27.2	26.1	22.3	16.8	12.2
		MCDB	10.1	11.7	15.3	19.5	22.9	27.6	30.3	30.4	28.8	24.7	18.9	14.5
	5%	WB	7.3	9.1	11.8	16.0	19.4	23.7	26.5	26.7	25.4	21.1	15.7	10.9
		MCDB	9.4	10.8	14.1	18.4	22.2	26.1	29.6	29.8	27.9	23.5	18.0	13.3
	10%	WB	6.3	7.6	10.6	15.1	18.6	22.6	26.0	26.4	24.7	19.8	14.9	9.9
		MCDB	8.8	9.8	12.8	17.4	21.3	25.0	28.8	29.4	27.2	22.4	17.5	12.5
Mean Daily Temperature Range	5% DB	MDBR	5.0	5.5	6.3	6.6	6.3	5.0	4.8	5.1	5.1	5.3	5.5	5.1
		MCDBR	5.9	6.8	8.3	8.1	7.7	6.2	6.1	6.0	5.7	5.8	6.2	6.1
		MCWBR	4.4	5.1	6.0	5.1	4.4	3.2	2.9	2.7	3.0	3.8	4.5	4.8
	5% WB	MCDBR	5.4	5.6	7.5	6.8	6.3	5.4	5.6	5.4	5.1	4.9	5.2	5.6
		MCWBR	4.5	4.9	6.3	4.8	4.2	2.8	2.9	2.7	3.1	3.9	4.6	5.3
Clear-Sky Solar Irradiance	taub	0.390	0.431	0.493	0.536	0.544	0.568	0.561	0.533	0.494	0.462	0.416	0.389	
	taud	2.197	2.095	1.944	1.888	1.938	1.950	2.020	2.085	2.162	2.171	2.208	2.216	
	Ebn at Noon	787	795	776	765	764	744	746	759	769	756	753	761	
	Edh at Noon	113	140	177	196	189	186	173	159	141	129	112	105	
All-Sky Solar Radiation	RadAvg	2.46	3.13	4.08	4.82	5.21	4.67	4.98	5.12	4.03	3.24	2.58	2.21	
	RadStd	0.16	0.24	0.28	0.47	0.57	0.44	0.67	0.46	0.37	0.33	0.23	0.16	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page